MTH207

Examination - July Semester 2025

Linear Algebra

Wednesday, 12 November 2025 04:00 pm - 06:00 pm

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **SIX (6)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

A chef has to prepare a feast comprising a number of vegetable, meat and dessert dishes. He has 60 min to perform the task and is given a budget of $51. The chef must personally make the dishes one at a time. He also has 32 interns, and each intern can help him with one dish only.

Each type of dish takes up some of the chef's time, incurs some cost, and requires a few interns to help according to the table below:

	Time (min)	**Cost ($)**	**No. of Interns**
Vegetable dish	5	2	0
Meat dish	8	5	2
Dessert dish	6	3	1

Question 1a

Suppose the chef makes use of exactly all the given time, budget and number of interns. Let v, m and d be the number of vegetable, meat, and dessert dishes respectively. Write down a system of linear equations that v, m and d must satisfy.

(6 marks)

Question 1b

The system of linear equations in Question 1a can be cast as a matrix equation below:

$$A \begin{pmatrix} v \\ m \\ d \end{pmatrix} = \mathbf{b},$$

where A is a 3-by-3 matrix and $\mathbf{b}$ is a 3-by-1 column vector. Write down A and $\mathbf{b}$ explicitly.

(2 marks)

Question 1c

Calculate the determinant of matrix A.

(2 marks)

Question 1d

From your answer to Question 1c, explain whether it is possible to solve the system of linear equations in Question 1a or not. If such a solution exists, **use the inverse of** A to solve for v, m and d. If a solution is not possible or is otherwise not acceptable, explain why not.

(10 marks)

Question 2

A linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ is represented by the matrix below:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 0 & -4 \end{pmatrix}.$$

Question 2a

From the size of matrix A, state and explain the maximum possible rank of A. Use the rank-nullity theorem to infer the minimum dimension of the null space of A.

(4 marks)

Question 2b

Compute the reduced row echelon form (RREF) of A and state the rank of A.

(5 marks)

Question 2c

Using the RREF in Question 2b, write down the column space and row space of A.

(6 marks)

Question 2d

Using the RREF in Question 2b, solve for the null space of A.

(5 marks)

Question 3

Consider the two sets of vectors in $\mathbb{R}^3$ below:

$$U = \{\mathbf{u_1}, \quad \mathbf{u_2}, \quad \mathbf{u_3}\} \text{ where } \mathbf{u_1} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \mathbf{u_2} = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}, \mathbf{u_3} = \begin{pmatrix} 3 \\ -1 \\ -1 \end{pmatrix};$$

$$V = \{\mathbf{v_1}, \quad \mathbf{v_2}, \quad \mathbf{v_3}\} \text{ where } \mathbf{v_1} = \begin{pmatrix} 5 \\ 1 \\ 5 \end{pmatrix}, \mathbf{v_2} = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}, \mathbf{v_3} = \begin{pmatrix} 4 \\ 0 \\ 7 \end{pmatrix}.$$

Question 3a

Determine whether U is a linearly independent set.

(4 marks)

Question 3b

Apply the Gram-Schmidt process to obtain an orthonormal basis $\{\mathbf{w_1}, \mathbf{w_2}\}$ for $Span\,\{\mathbf{u_1}, \mathbf{u_2}\}$. You may assume that $\{\mathbf{u_1}, \mathbf{u_2}\}$ is a linearly independent set.

(10 marks)

Question 3c

Show that V spans $\mathbb{R}^3$.

(6 marks)

Question 4

Let $A = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 8 & 0 & 0 \end{pmatrix}$.

Consider the eigenvalue equation, $A\mathbf{x} = \lambda\mathbf{x}$, where $\mathbf{x}$ is an eigenvector belonging to the eigenvalue λ.

Question 4a

From the eigenvalue equation, derive the characteristic equation below while **making clear explicit mathematical arguments**.

$$|A - \lambda I| = 0$$

Hence, show that the eigenvalues of A are 2, 4, and -4.

(6 marks)

Question 4b

Calculate the eigenvectors $\mathbf{x_2}$, $\mathbf{x_4}$ and $\mathbf{x_{-4}}$ belonging respectively to eigenvalues 2, 4, and -4.

(9 marks)

Question 4c

The matrix A can be diagonalised as follows:

$$A = QDQ^{-1}.$$

Show **explicitly** that the columns of Q are eigenvectors of A and the diagonal elements of D are the respective eigenvalues of A in corresponding order.

(5 marks)

Question 5

Let T be a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Question 5a

Suppose T is one-to-one. Show this is equivalent to the statement that for $\mathbf{v} \in \mathbb{R}^n$, $T(\mathbf{v}) = \mathbf{0}$ has only the trivial solution $\mathbf{v} = \mathbf{0}$.

(10 marks)

Question 5b

Suppose $T : \mathbb{R}^3 \to \mathbb{R}^3$. For $\mathbf{x} \in \mathbb{R}^3$, $T(\mathbf{x}) = A\mathbf{x}$ such that

$$A = \begin{pmatrix} 2 & 5 & 3 \\ 1 & 6 & 2 \\ 4 & s & 7 \end{pmatrix}$$

where $s \in \mathbb{R}$. For the whole of Question 5b, assume that T is **NOT** onto.

Question 5b(i)

Solve for s.

(4 marks)

Question 5b(ii)

Find the row echelon form (REF) of the augmented matrix $[A|\mathbf{y}]$ of the linear system $A\mathbf{x} = \mathbf{y}$, where $\mathbf{y} = (a\ b\ c)^T \in \mathbb{R}^3$. With the help of this REF, define a subspace $W \subset \mathbb{R}^3$ such that $T_W : \mathbb{R}^3 \to W$ is onto, where $T_W(\mathbf{x}) = T(\mathbf{x}), \mathbf{x} \in \mathbb{R}^3$.

(6 marks)

----- END OF PAPER -----